Luc A. **Barrett**

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Experimental physics student intersted in the intersections of quantum information, particle physics, and cosmology

Education

Cornell University — *Ithaca, NY*
Doctor of Philosophy in Applied Physics — *2025 - Present*

University of Massachusetts Amherst — *Amherst, MA*
Master of Science in Computer Science — *2024 - 2025*

Bay State Fellow
Relevant Courses: Reinforcement Learning, Machine Learning, Quantum Communication, Advanced Robot Dynamics & Control

University of Massachusetts Amherst — *Amherst, MA*
Bachelors of Science in Physics, *Cum Laude* — *2020 - 2024*
Commonwealth Honors Scholar with Greatest Distinction

GPA: 3.94
Thesis: *"Characterization of Drifting Charge Clusters in Liquid Xenon Produced by a Laser-Driven Photocathode for the nEXO Experiment"*, Advised by Prof. Krishna Kumar, Dr. Jason Bane

University of Massachusetts Amherst — *Amherst, MA*
Bachelors of Science in Mathematics, *Cum Laude* — *2020 - 2024*

Relevant Courses: Nonlinear Dynamics & Chaos, Differential Geometry, Real Analysis, Partial Differential Equations

University of Massachusetts, Amherst — *Amherst, MA*
Bachelors of Science in Computer Science, *Cum Laude* — *2020 - 2024*

Relevant Courses: Intro. to Machine Learning, Natural Language Processing, Artificial Intelligence, Quantum Algorithms

Research Experience

UMass Amherst College of Information & Computer Sciences — *Amherst, MA*
Master's Project — *2024 - 2025*

- Apply modern reinforcement learning algorithms to optimize link generation between quantum memory registers with various noise models, and minimal simplifying assumptions

Simulation | Quantum Information

UMass Amherst — *Amherst, MA*
Undergraduate Research — *2021 - 2024*

- Worked on the MOLLER experiment, developing a software tool to check 'two-bounce' conditions. Validates the design of the collimator system and ensures no accidental avenues for noise
  - Used Rust, multithreading, and optimized collision check algorithms to bring simulation time from several days to a couple of minutes. Uses geometry directly imported from Geant4.
  - Presented in APS DNP CEU undergraduate poster session
- Worked on the nEXO neutrinoless double beta decay ($0\nu\beta\beta$) experiment
  - Under Prof. Krishna Kumar, worked on characterization of charge clusters drifting in liquid xenon produced by a laser-driven photocathode
    - Work from the team presented in poster session at Neutrino 2024
  - Under Prof. Andrea Pocar, worked on studies using experimental data and GPU-accelerated Chroma simulations to study optical properties of materials in LXe, at the scintillation wavelength of Xe
    - Work presented in internal nEXO Collaboration meeting
- Started development of a Julia simulation package to simulate gaussian quantum states. Implement efficient representations and operations to study these states, with applications to quantum sensing and algorithms

Nuclear Physics | $0\nu\beta\beta$ | Algorithm Design | Simulation | Quantum Information

Posters & Presentations

Chroma Optics Measurements — Oct 2024
L. Barrett (Speaker) — nEXO Light Simulation Workshop, McGill University, Québec, CA

Photo-Induced Charge Calibration for nEXO — Jun 2024
D. Cesmecioglu (Presenter), J. Bane, L. Barrett, K.S. Kumar, M. Loscar, A. Nolan, J. Zhu — Neutrino 2024, Milan, IT

Program to Identify Secondary Background Sources in the MOLLER Experiment — Nov 2023
L. Barrett (Presenter), J. Mott, K.S. Kumar — APS DNP 2023, Waikoloa, HI, US

Awards & Honors

2025 **Fulbright Semifinalist**, Fulbright Association (Research/Study, The Netherlands)

2024 **Bay State Fellowship**, UMass Amherst College of Information & Computer Sciences
∟ *Competitive fellowship providing a tuition waiver and stipend for an accelerated M.S. Computer Science degree*

2024 **Kandula Sastry Undergraduate Award**, UMass Amherst Department of Physics
∟ *Awarded annually to the outstanding physics student in the graduating class*

2024 **Phi Beta Kappa Membership**, $\Phi\mathrm{BK}$ Society

2023 **LeRoy F. Cook Jr. Memorial Award**, UMass Amherst Department of Physics
∟ *Awarded annually recognizing academic excellence and involvement in teaching or outreach*

2023 **Phi Kappa Phi Membership**, $\Phi\mathrm{K}\Phi$ Society

2024 **Dean's List**, UMass Amherst
∟ *Awarded 7 times, all of my full-time semesters, for achieving a semester GPA of 3.5+*

Teaching Experience

Higher Education

Cornell University College of Applied & Engineering Physics — *Ithaca, NY*
Graduate Teaching Assistant — *2025-2026*

UMass Amherst College of Information & Computer Sciences — *Amherst, MA*
Graduate Teaching Assistant — *2024 - 2025*

- CS648: Quantum Information Systems. For the class of ~40 graduate students, I held weekly office hours, and additional review lectures to cover optional practice problem sets I designed. Graded and proctored midterm & final exams
- CS250: Introduction to Computation. For the class of ~300 undergraduate students, I held weekly office hours, proctored exams, and graded weekly homework assignments.

UMass Amherst — *Amherst, MA*
Undergraduate Teaching Assistant — *2024 - 2025*

- CS490Q: Quantum Information Science. Held in-person office hours and exam review. Graded homework assignments
- PHYS281: Computational Physics (x2). Assisted students with in-class exercises and held office hours. Held optional supplemental lectures to cover additional relevant content (OOP, data structures, API's, multiprocessing, cluster computing).
- PHYS181: Introduction to Mechanics. Assisted students with assignments in-class, held exam-review and office hours.

Secondary Education

Hotchkiss School — *Ithaca, NY*
Instructor — *June 2025*

- Instructor for inaugural 'Euler-to-Einstein' mathematics program for motivated high school students
- Held classes covering advanced math topics accessible to students who have taken up to AP Calculus
- Designed curriculum components covering multivariable calculus and calculus of variations

Mentoring Experience

UMass Amherst — *Amherst, MA*
Undergraduate Peer Mentor — *2022 - 2024*

- Invited by both physics and CS department faculty to be paired with incoming freshman students as a peer mentor. Met regularly to provide social and academic support, particularly when involving course selection/planning, and getting involved in research (both with faculty on campus and applying to REUs)